GATE DA Notes — Machine Learning and AI

Machine Learning

Supervised learning basics

- **Linear regression:** minimise squared error. Normal equation: $w = (X^TX)^{-1}X^Ty$. Gradient step: $w \leftarrow w - \eta \cdot \nabla L$.
- **Ridge regression:** minimise $\|y - Xw\|^2 + \lambda\|w\|^2$. Solution $w = (X^TX + \lambda I)^{-1}X^Ty$. It shrinks weights (never exactly zero), always invertible for $\lambda > 0$, increases bias and lowers variance as λ grows.
- **Logistic regression:** $P(y=1|x) = \sigma(w^Tx)$, $\sigma(z) = 1/(1+e^{-z})$. Trained by maximising likelihood (cross-entropy loss). Decision boundary is linear. It is a discriminative model.
- **k-NN:** predict by majority (or mean) of the k closest points. Small k: low bias, high variance. Needs feature scaling. No training phase.
- **Naive Bayes:** assumes features are conditionally independent given the class; $P(y|x) \propto P(y)\prod P(x_i|y)$. It is a generative model. For K binary features and 2 classes: parameters = 1 (prior) + 2K ($P(x_i=1|y)$) = 2K + 1.
- **LDA (linear discriminant analysis):** assumes Gaussian classes with a shared covariance; gives a linear boundary. Fisher criterion maximises between-class scatter over within-class scatter: max $w^TS_B\, w$ / $w^TS_W\, w$.
- **SVM:** maximum-margin classifier. Margin = $2/\|w\|$. Only support vectors determine the boundary. Soft margin uses C to trade margin against errors.
- **Decision trees:** split on the attribute with highest information gain. Entropy $H = -\Sigma\, p_i \log_2 p_i$. $IG = H(\text{parent}) - \Sigma\, (n_k/n)\cdot H(\text{child}_k)$. Gini $= 1 - \Sigma\, p_i^2$.

Model selection

- **Bias–variance:** error = bias² + variance + noise. Complex models: low bias, high variance (overfitting).
- **k-fold CV:** split into k folds; train on k−1, validate on 1; repeat k times. **Leave-one-out CV:** k = n, so n validation splits for n samples.
- Always count the splits on the data used for model selection only (not the held-out test set).

Neural networks

- A neuron computes $a = f(w^Tx + b)$. Common activations: sigmoid, tanh, ReLU = max(0, z).
- Parameters of a fully connected layer with m inputs and n outputs: m·n + n (with biases).
- Forward pass is matrix multiplication plus activation; backpropagation applies the chain rule layer by layer.

- Without a non-linear activation, stacked layers collapse to one linear map.

Unsupervised learning

- **k-means:** repeat (assign each point to nearest centroid; recompute centroids as means). Minimises within-cluster squared distance; converges to a local minimum; sensitive to initialisation. k-medoid uses actual points as centres.
- **Hierarchical clustering:** bottom-up (agglomerative) merges closest clusters; top-down (divisive) splits. Single linkage = minimum pairwise distance; complete linkage = maximum.
- **PCA:** centre data; take eigenvectors of the covariance matrix with the largest eigenvalues. Variance along a component = its eigenvalue. Reducing to k dimensions keeps the top k components; the principal components are orthogonal.

Quick traps

- Data scaling: z-score = (x − mean)/std; min-max = (x − min)/(max − min).
- Discriminative models (logistic regression, SVM) learn $P(y|x)$ or a boundary; generative models (naive Bayes, LDA) model $P(x|y)P(y)$.
- Regularisation lowers variance, raises bias.

Artificial Intelligence

Search

- **Uninformed:** BFS (complete, optimal for equal step costs, O(b^d) memory), DFS (O(bm) memory, not optimal), uniform-cost search (optimal for positive costs), depth-limited and iterative deepening.
- **Informed:** greedy best-first uses h(n); **A*** uses f(n) = g(n) + h(n). A* is optimal when h is admissible (never overestimates) for tree search, and consistent for graph search. If h_1 and h_2 are admissible, so are $\min(h_1, h_2)$ and any convex combination; $\max(h_1, h_2)$ is also admissible and dominates both.
- **Adversarial search:** minimax: MAX picks the child with the highest value, MIN the lowest. Alpha–beta pruning gives the same root value as minimax; best-case ordering examines about O(b^(d/2)) nodes.

Logic

- **Propositional:** truth tables, equivalences (De Morgan, contrapositive: $p \to q \equiv \neg q \to \neg p \equiv \neg p \vee q$). X entails Y if Y is true in every model where X is true. Tautology = always true; satisfiable = true in at least one model.
- **Predicate (first-order):** $\forall x\, P(x)$ and $\exists x\, P(x)$; $\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$; $\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$. Careful: $\forall x \exists y\, R(x,y)$ is not equivalent to $\exists y \forall x\, R(x,y)$; the second implies the first.

Reasoning under uncertainty

- **Bayesian network:** joint = $\prod P(X_i \mid \text{parents}(X_i))$. A network with n Boolean variables needs Σ 2^(#parents) numbers.
- **Conditional independence (d-separation):** chain A→B→C: $A \perp C \mid B$. Fork A←B→C: $A \perp C \mid B$. Collider A→B←C: $A \perp C$ marginally, but dependent once B (or a descendant) is observed.
- **Exact inference by variable elimination:** multiply the factors that mention a variable, sum it out, repeat; order affects cost.
- **Approximate inference by sampling:** prior sampling, rejection sampling (discard samples inconsistent with the evidence), likelihood weighting, Gibbs/MCMC. Estimates improve with more samples.

Revision checklist

- ☐ Ridge, logistic, naive Bayes parameter counts
- ☐ Entropy and information gain by hand
- ☐ LOOCV / k-fold split counts
- ☐ Parameters in a feed-forward net
- ☐ k-means iteration and PCA eigenvalue/variance questions
- ☐ A* admissibility, minimax and alpha–beta by hand
- ☐ d-separation in three structures
- ☐ Variable elimination on a small network